

Literature status: the $p, q > 2$ Lipschitz extension upper bound

Run `fa_banach_001`; source arXiv:1506.04398

Source question

Naor and Rabani's paper *On Lipschitz extension from finite subsets* defines $e_n(X, Z)$ as the supremal Lipschitz extension loss for Banach-space-valued functions from subsets of X of cardinality at most n . In Section 1.4 they survey open questions about $e_n(\ell_p, \ell_q)$. For $p, q \in (2, \infty)$, the only upper bound that they record is the general Lee–Naor finite-metric bound

$$e_n(\ell_p, \ell_q) \lesssim \frac{\log n}{\log \log n},$$

and they explicitly say that no asymptotically smaller upper bound was known.

By Ball's extension theorem [Bal92], combined with the fact [NPSS06] that ℓ_p has Markov type 2 when $p \in [2, \infty)$, if $1 < q \leq 2 \leq p < \infty$ then $e(\ell_p, \ell_q) \leq C(p, q)$ for some $C(p, q) \in (0, \infty)$. Here the asymptotic dependence of $C(p, q)$ as $p \rightarrow \infty$ or $q \rightarrow 1$ remains unknown. In particular, a famous and longstanding open question of Ball [Bal92] asks whether $e(\ell_2, \ell_1)$ is finite or infinite. The best known bounds on $e_n(\ell_p, \ell_q)$ for the remaining values of $p, q \in [1, \infty]$ are as follows.

$$p, q \in [1, 2] \text{ and } q \neq 1 \implies \left(\frac{\log n}{\log \log n} \right)^{\frac{1}{p} - \frac{1}{2}} \lesssim_q e_n(\ell_p, \ell_q) \lesssim_p (\log n)^{\frac{1}{p}}, \quad (18)$$

$$1 \leq p \leq 2 \implies \frac{(\log n)^{\frac{1}{p} - \frac{1}{2}}}{(\log \log n)^{\frac{1}{p}}} \lesssim e_n(\ell_p, \ell_1) \lesssim_p (\log n)^{\frac{1}{p}}. \quad (19)$$

$$p, q \in (2, \infty) \implies \left(\frac{\log n}{\log \log n} \right)^{\frac{q-2}{q^2}} \lesssim e_n(\ell_p, \ell_q) \lesssim \frac{\log n}{\log \log n}, \quad (20)$$

$$1 \leq p \leq 2 \leq q < \infty \implies \left(\frac{\log n}{\log \log n} \right)^{\max\{\frac{1}{p} - \frac{1}{2}, \frac{q-p}{q^2}\}} \lesssim_q e_n(\ell_p, \ell_q) \lesssim_p (\log n)^{\frac{1}{p}}. \quad (21)$$

The right hand inequalities in (18), (19), (21) are due to [LN05]. More generally, it was shown in [LN05] that for $p \in (1, 2]$ we have $e_n(\ell_p, Z) \lesssim_p (\log n)^{1/p}$ for every Banach space Z . It would be interesting to determine the best exponent of $\log n$ in this context of general Banach space targets, as well as when the target Z is allowed to range over some special classes of Banach spaces (e.g., for Banach spaces of cotype 2 it is known [MN13a] that this exponent cannot be smaller than $1/4$).

No upper bound on $e_n(\ell_p, \ell_q)$ that is asymptotically smaller than the general upper bound [LN05] of $O((\log n)/\log \log n)$ is known when $p, q \in (2, \infty)$, and similarly for $e_n(\ell_1, \ell_q)$ for $q \in (2, \infty)$.

Let $(Z, \|\cdot\|_Z)$ be a Banach space that has Rademacher type p for some $p > 1$. By a theorem

Figure 1: arXiv:1506.04398, page 8: the source passage recording the $p, q > 2$ upper-bound gap for $e_n(\ell_p, \ell_q)$.

Supporting literature status

Naor and Ren's later paper *Euclidean embedding, randomized clustering, and Lipschitz extension for finite and doubling subsets of L_p when $p > 2$* proves a stronger Banach-target extension theorem for domains L_p . They also explicitly state that the theorem answers a natural question left open by [1] and stated in [2].

The following Lipschitz extension theorem treats the analog of (1.9) in the previously unknown range $p > 2$:

Theorem 1.5. *If $2 < p < \infty$ and $n \in \{2, 3, \dots\}$, then*

$$e_n(L_p) \lesssim p^2 \sqrt{\log n}.$$

Theorem 1.5 answers the natural question that was left open by [53] (and stated elsewhere, e.g. [82]) whether for every (indeed, any) fixed $2 < p < \infty$ we have

$$(1.10) \quad \lim_{n \rightarrow \infty} e_n(L_p) \frac{\log \log n}{\log n} = 0.$$

In other words, prior to Theorem 1.5 it was not even known if one could provide an upper bound on $e_n(L_p)$ that is asymptotically better than merely using the fact that L_p is a metric space through (1.8).

Figure 2: arXiv:2502.10543, page 7: Theorem 1.5 and the explicit answer statement.

- [81] A. Naor, Y. Peres, O. Schramm, and S. Sheffield. Markov chains in smooth Banach spaces and Gromov-hyperbolic metric spaces. *Duke Math. J.*, 134(1):165–197, 2006.
- [82] A. Naor and Y. Rabani. On Lipschitz extension from finite subsets. *Israel J. Math.*, 219(1):115–161, 2017.
- [83] A. Naor and K. Ren. ℓ_p has nontrivial Euclidean distortion growth when $2 < p < 4$. Preprint available at <https://arxiv.org/pdf/2410.21931v1> (observe that this is version v1). 2025

Figure 3: arXiv:2502.10543, page 59: bibliography entry [82], identifying [2] as Naor–Rabani’s *On Lipschitz extension from finite subsets*.

Proof intuition

The source passage asks whether finite subsets of ℓ_p , with $p > 2$, have any extension behavior better than arbitrary finite metric spaces when the target is another ℓ_q , $q > 2$. Naor–Ren prove something stronger at the domain level: every n -point subset of L_p has Banach-valued Lipschitz extension loss $O_p(\sqrt{\log n})$. Since ℓ_p embeds isometrically into an L_p space, an extension theorem on L_p immediately restricts to ℓ_p . Then choosing the target Banach space to be ℓ_q gives exactly the source paper’s $p, q > 2$ case.

The literature answer

Theorem 1. *For every fixed $p, q \in (2, \infty)$,*

$$e_n(\ell_p, \ell_q) \lesssim_p \sqrt{\log n}.$$

In particular,

$$e_n(\ell_p, \ell_q) = o\left(\frac{\log n}{\log \log n}\right) \quad (n \rightarrow \infty).$$

Thus the $p, q > 2$ upper-bound gap recorded in [2] has been answered by [3].

Proof. Naor–Ren’s Theorem 1.5 [3] states that if $2 < p < \infty$, then

$$e_n(L_p) \lesssim p^2 \sqrt{\log n}.$$

Here $e_n(L_p)$ is the supremum over all subsets of L_p with cardinality at most n and over all Banach targets.

Embed ℓ_p isometrically into an L_p space; for instance, use disjoint measurable sets and represent a sequence by the corresponding step function after the standard normalization. Let $S \subset \ell_p$ with $|S| \leq n$, and let $f : S \rightarrow \ell_q$ be Lipschitz. Viewing S as a subset of L_p , Theorem 1.5 of [3] extends f to a map $F : L_p \rightarrow \ell_q$ with

$$\|F\|_{\text{Lip}} \lesssim p^2 \sqrt{\log n} \|f\|_{\text{Lip}}.$$

Restricting F back to the embedded copy of ℓ_p gives an extension of f to ℓ_p with the same Lipschitz bound. Taking the supremum over S and f yields

$$e_n(\ell_p, \ell_q) \lesssim p^2 \sqrt{\log n}.$$

For fixed p , $p^2 \sqrt{\log n} = o((\log n)/\log \log n)$, so this is asymptotically smaller than the general [1] bound recorded in the source paper. $\square$

Verification notes and limitations

This packet is a literature-status identification, not a new proof from the run. The supporting paper explicitly says that Theorem 1.5 answers the natural question left open by [1] and stated in [2]; its bibliography identifies [2] as Naor–Rabani’s 2017 journal version of arXiv:1506.04398.

The packet addresses the source paper’s $p, q > 2$ upper-bound gap for $e_n(\ell_p, \ell_q)$. It does not answer the separate Kalton Holder-extension questions or the finite-dimensional Ball extension question that also appear in arXiv:1506.04398. It also does not assert sharp asymptotics for $e_n(\ell_p, \ell_q)$; it records that an upper bound strictly below $O((\log n)/\log \log n)$ is now known.

Before promotion, the run lightweight indexes were searched for arXiv:1506.04398, the title phrase, “Lipschitz extension”, “finite subsets”, “ell_p”, “ell_q”, and nearby core terms. No existing exact packet or attempt was found. Broad hits to other Lipschitz/free-space packets were not duplicates, and human-only archive contents were not opened.

Human review recommendation

Treat the $p, q > 2$ upper-bound gap recorded in arXiv:1506.04398 Section 1.4 as already answered by arXiv:2502.10543. Keep the named Kalton and Ball questions from the same source paper separate.

References

- [1] J. R. Lee and A. Naor, *Extending Lipschitz functions via random metric partitions*, Invent. Math. 160 (2005), 59–95.
- [2] A. Naor and Y. Rabani, *On Lipschitz extension from finite subsets*, Israel J. Math. 219 (2017), 115–161; arXiv:1506.04398.
- [3] A. Naor and K. Ren, *Euclidean embedding, randomized clustering, and Lipschitz extension for finite and doubling subsets of L_p when $p > 2$* , arXiv:2502.10543.